

Profile

A mathematically highly skilled R&D engineer in satellite communications with a pioneering publication record in stochastic geometry modeling of satellite communication systems, including a first-author publication in the high-impact journal *Institute of Electrical and Electronics Engineers (IEEE) Transactions on Communications (TCOM)* and a first-author presentation in the flagship conference *IEEE GLOBECOM*. A creative mindset that seeks simplicity – not complexity – and who is not afraid to fiddle with the details. Committed to contributing to the growing satellite systems industry, bringing my unique expertise in a collaborative, challenging, development-driven environment. Always striving to grow professionally and spiritually, and to cultivate compassion and humility in all aspects of life. Ready to relocate; FI/EU citizen.

Skills

• Data analysis and visualization • Inter-institute collaboration • Research • Stakeholder management • Writing
• C, MATLAB, PYTHON • Inter-satellite data routing and performance optimization • Simulation of satellite constellations •
Space-ground link performance modeling and optimization in satellite communications • Stochastic processes

Experience and Impact

5/2025- **Senior Researcher**, *Aalto University*, School of Electrical Engineering, Espoo, Finland

- Managed 2 R&D projects as the corresponding author and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication.
- Mentored junior researchers as a bachelor's degree supervisor and helped to establish future talents.
- Voluntary work as a peer reviewer in various journals, including *IEEE TCOM*, *IEEE TWC*, *IEEE WCL*, *IEEE COMML*, and *IEEE TMC*.
- Proposed a novel closed-form Gaussian autocorrelation function for the aggregate interference at a low Earth orbit (LEO) satellite terminal from Poisson-distributed terrestrial signal terminals, applicable, *e.g.*, in machine learning (ML) algorithms for satellite communications as a tractable hypothesis class.
- Proposed, analyzed, and simulated successive interference cancellation (SIC) algorithms for multibeam LEO uplink, improving coverage performance by 900% for the third-strongest Earth transmitter in terms of signal-to-interference ratio (SIR). The SIC improves user fairness and enables serving multiple terrestrial terminals per cell, while preserving the average total network throughput at the LEO satellites, which is optimized around 1 (bit/s/Hz).

8/2023- **Research Internship**, *University of Notre Dame*, Dept. of Electrical Engineering, South Bend, IN, US

- 2/2024
- Based on rigorous mathematical analysis, a novel statistical SIR model was proposed by characterizing the SIR at a LEO satellite receiver using a Lomax distribution: a well-known heavy-tailed distribution. Consequently, many results for the Lomax distribution can be exploited in performance analysis and optimization of LEO communications.
 - Coordinated a distributed team in a cross-functional project and handled the full delivery pipeline (simulation, analysis, documentation, peer review) to publication in the high-impact journal *IEEE TCOM*.
 - Identified an optimal cell size in a LEO uplink that maximizes average throughput and quantified a trade-off between average throughput and performance consistency: -10 dB cell size provides high average throughput but highly varying link quality of service (QoS) over the beams, whereas -3 dB cells provide more stable QoS but are detrimental to the throughput.
 - International relations.
 - Secured project financing of 4349 € for personal internship expenses from the Väisälä Foundation.

05/2021- **PhD Researcher**, *Aalto University*, Signal Processing of Wireless Networks, Espoo, Finland

- 05/2025
- Built, maintained, and documented a multi-language simulation stack (GIT, MATLAB, MATHEMATICA, OCTAVE, PYTHON) including algorithms for analysis, simulation, and visualization, used across several research projects.
 - Developed a novel and well-cited stochastic geometry framework for the analysis of LEO networks, reducing simulation time by an order of magnitude 100 times more computationally efficient than the MATLAB Satellite Communication Toolbox, while providing equivalent results on the network performance metrics.
 - Industry presentation at *IEEE GLOBECOM* on a novel stochastic geometry system model for LEO uplink performance metrics, characterizing the SIR with a gamma distribution. Consequently, many results for the gamma distribution can be exploited in performance analysis and optimization of LEO communications.
 - Proposed, analyzed, and simulated a minimal-overhead slotted ALOHA-like inter-satellite link (ISL) data routing algorithm in a constellation of 10 planes, 6 satellites in each plane at the orbital altitude of 1000 km, using graph algorithms. The QoS was optimized at an average latency of 30 ms and total network spectral efficiency 1 (bit/s/Hz).
 - Proposed, analyzed, and simulated a minimal-overhead, grant-free, random medium access control (MAC) for multibeam LEO uplink that mitigates the high variation in individual link performance while preserving average network throughput, optimized around 1 (bit/s/Hz).

Education

Expected 2026 **Doctor of Science, PhD**, *Aalto University, School of Electrical Engineering*, Signal Processing for Communications, • Stochastic Geometry Analysis and Simulation of LEO networks
MSc, *Univ. of Helsinki*, Applied Analysis, • Finite Element Method (FEM) • Antenna Theory
BSc, *Univ. of Helsinki*, Applied Analysis, • Optimal Control Theory • System Performance Optimization

References

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- MSc. Abid Afridi, Aalto University, abid.afridi@aalto.fi